4. Travelling Salesman Problem - Local Beam Search

Python practical code

```
dist = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

beam = [[0, 1, 2, 3, 0], [0, 2, 1, 3, 0]]

def cost(r):
    total = 0
    for i in range(4):
        total += dist[r[i]][r[i + 1]]
    return total

for _ in range(5):
    new = beam.copy()

    for r in beam:
        for i in range(1, 4):
            for j in range(i + 1, 4):
                route = r.copy()
                route[i], route[j] = route[j], route[i]

                if route not in new:
                    new.append(route)

    new.sort(key=cost)
    beam = new[:2]

best = min(beam, key=cost)

print("Best Route:", best)
print("Distance:", cost(best))
```